

Report - Create 3D points from cast shadows on a 3D object

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First of all, I needed to be able to recognize the red shadow cast on the statue and the wireframes. To do that, I created a binary image by subtracting the blue channel color from the red channel color, and if the difference was greater than 15, I considered that the color was more red than blue.

Now, I have an image full of zero values, and the only non-zero values correspond to the red shadow. I need to perform homography on each picture, and to do that I need 4 points per image. The 4 points used for homography correspond to the 4 regions of red shadow casted onto the wireframe.

To find the y coordinates, I create a list indicating whenever a row of the image has a red pixel or not. So, whenever there is a change from region of 'row with no red pixel' to 'row with red pixel', and vice versa, I store that position in a list. So, essentially I know when the 'chunks' of red shadow begin and end in the image. I know that the first 4 elements in this list correspond to the 2 upper wireframes, and the last 4 elements to the 2 lower wireframes.

To get the x coordinates, I just need to position into the rows I know there are red pixels, and find the index of the first appearance of a red shadow.

Now that I can extract the position of the red shadow cast onto the wireframe, I can use those points to perform homography.

From the homographed images, I need to get xyz coordinates. To do that, I need to choose only 1 red pixel point per row of the image, so, I choose the first instance of the highest value per row. But before calculating the xyz coordinates, I mask over the first and last 5 rows, and first and last 5 columns of the image, to eliminate any pixels left of the red shadow casted into the wireframe.

Finally, I am able to calculate the correct xyz coordinates for the points in a 3d world for each image, and save the results into a F11115117.xyz file. There are 4046 points in the F11115117.xyz file.

By using CloudCompare with the files F11115117.xyz and the GroundTruth.stl, I can analyze that the average distance between the points in my F11115117.xyz file and the GroundTruth.stl is of 1.7mm, and the maximum distance between them is of 5.17mm.

1	Min dist.	0
2	Max dist.	5.17137
3	Avg dist.	1.70065
4	Sigma	1.23967
5	Max error	0.480149

This is the comparison of GroundTruth.stl (to the left), and F11115117.xyz (to the right) side by side.

